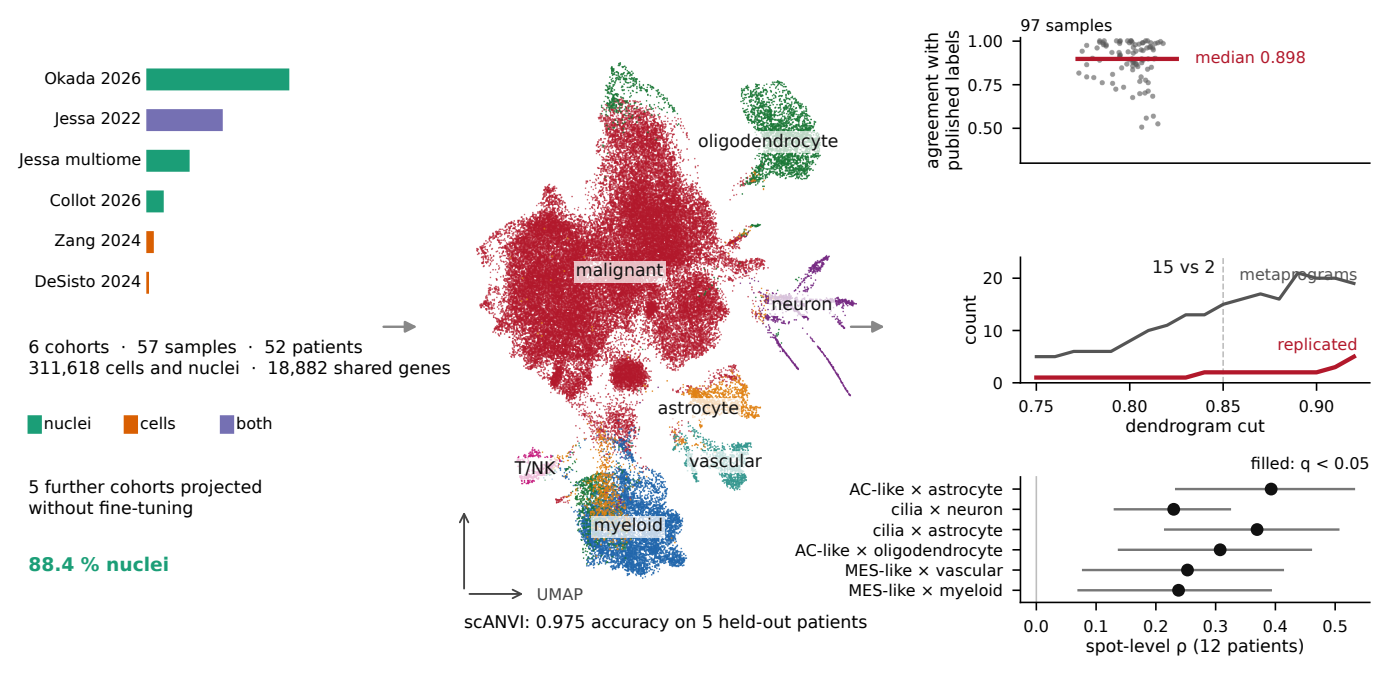

Okada 2026
Jessa 2022
Jessa multiome
Collot 2026
Zang 2024
DeSisto 2024
6 cohorts · 57 samples · 52 patients
311,618 cells and nuclei · 18,882 shared genes
nuclei
cells
both
5 further cohorts projected
without fine-tuning
88.4 % nuclei
oligodendrocyte
malignant
neuron
astrocyte
vascular
T/NK
myeloid
UMAP
scANVI: 0.975 accuracy on 5 held-out patients
97 samples
agreement with
published labels
1.00
0.75
0.50
median 0.898
15 vs 2
metaprograms
replicated
count
20
10
0
0.75
0.80
0.85
0.90
dendrogram cut
filled: q < 0.05
AC-like × astrocyte
cilia × neuron
cilia × astrocyte
AC-like × oligodendrocyte
MES-like × vascular
MES-like × myeloid
0.0
0.1
0.2
0.3
0.4
0.5
spot-level ρ (12 patients)